

Theory of Epistemic Control / Revision-Robust Epistemic Transition Integrity

Status: WORKING TECHNICAL PROGRAM

Research question

What information must a bounded human–AI or autonomous research system retain so that a later correction can propagate through materially dependent claims, actions, and completion states—without leaving stale conclusions active or reopening unaffected work unnecessarily? RR-ETI turns that governance problem into a technical question about revision-aware state compression and cost-sensitive repair.

Contribution and current result

The program's strongest results are exact within a finite additive model. They include a Bayes-optimal posterior-impact threshold for coordinatewise repair; an exact conditional Jensen-gap expression for representation cost; a coarsest posterior-impact statistic under finite common-support and positive-cost assumptions; and an encoder–decoder regret decomposition. A finite-cost closure signature is sufficient for a fixed cost set. Under a fixed scalar closure objective, nonnegative costs, and a nesting-compatible canonical cut convention, the selected optimal repair sets form a nested chain with at most n strict exits and $n+1$ distinct canonical sets.

The larger closure-constrained compression problem remains open. The current manuscript therefore presents a scoped technical core plus an operational runtime for separating generation, verification, authorization, execution, publication, and completion. Naturalistic mathematical campaigns show that the runtime can preserve claim status, corrections, dependencies, and target continuity; they do not show that it outperforms simpler systems.

Methods

RR-ETI combines finite Bayesian decision models, noisy correction channels, typed provenance, dependency and closure graphs, asymmetric repair costs, maximum-weight closure/min-cut machinery, and versioned claim/status ledgers. Candidate results are paired with assumption tables, explicit falsifiers, negative-result retention, and separate validation axes for mathematical validity, computation, provenance, prior art, formalization, and external review.

Zackary's role

Zackary serves as the human program owner and research director: he framed the correction-integrity problem, maintained the target and governance boundary, directed the iterative technical program, and reserved authorship, legal, publication, and public-release decisions for human confirmation. The current evidence does not resolve final scholarly authorship or affiliation metadata.

AI and tool role

AI systems substantially assisted with model development, theorem exploration, literature synthesis, counterexample-oriented critique, workflow instrumentation, and manuscript drafting. Version control, structured ledgers, deterministic checks, and exact optimization tools supported traceability. Agreement among models is treated as internal evidence, not independent external review.

Verification

The finite-model results are internally coherent in the working manuscript, but no external claim-specific mathematical audit or load-bearing proof-assistant package was located. A cited 512-policy validator was not recovered as a complete publication packet. Operational demonstrations are naturalistic process evidence only. The planned comparative test is Study 14.1, which has not run.

Exact nonclaims

RR-ETI is not claimed to be a universal law of intelligence, a theory outside decision theory, or a demonstrated cause of better research outcomes. Finite-model results do not establish a continuous-cost closure-constrained correspondence. No causal claim is made that ETI outperforms full context, status labels alone, semantic retrieval, or any other baseline.

Public artifacts

None yet. No public RR-ETI repository or approved public manuscript was found during the bounded GitHub/web discovery.

Employer relevance

The program demonstrates work at the intersection of decision theory, provenance, knowledge-state design, correction handling, human–AI systems, and research governance. It is most relevant to roles involving agent reliability, evaluation design, auditable AI workflows, knowledge systems, or research operations where preserving scope and revisability matters as much as generating an answer.